Span and Linear Independence Reading

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In this lesson we extend the concepts of span and linear independence from Euclidean space to vector spaces. The definitions here are similar to those for Euclidean space, so lean on your previous knowledge.

Span

Just as in $\mathbf{R}^n$, a linear combination of a set of vectors $\{\mathbf{v}_1,\ \mathbf{v}_2, \ldots,\ \mathbf{v}_m\}$ is a sum of the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_m\mathbf{v}_m \qquad (1)$$

where $c_1,\ c_2, \ldots,\ c_m$ are real numbers. For instance, for the vectors $f(x) = \cos(x)$ and $g(x) = \log(x)$ in $C\ [1,\ 5]$, one possible linear combination is

$$7\cos(x) - \pi\log(x)$$

As in Euclidean space, the span of a set is defined in terms of linear combinations.

DEFINITION 5

Let $\mathcal{V} = \{\mathbf{v}_1,\ \mathbf{v}_2, \ldots,\ \mathbf{v}_m\}$ be a nonempty set of vectors in a vector space V. The span of this set is denoted $\text{span}\ \{\mathbf{v}_1,\ \mathbf{v}_2, \ldots,\ \mathbf{v}_m\}$ and is defined to be the set of all linear combinations of the form

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \ldots + c_m\mathbf{v}_m$$

where $c_1, c_2, \ldots, c_m$ can be any real numbers.

If $\mathcal{V}$ consists of infinitely many vectors, then we define $\text{span}(\mathcal{V})$ to be the set of all linear combinations of *finite* subsets of $\mathcal{V}$.

Example 1

Let $S = \text{span}\left\{x^2 - 2x + 3,\ -2x^2 + 3x + 1\right\}$ be a subset of $\mathbf{P}^2$. Determine if $p(x) = 10x^2 - 17x + 9$ is in S.

Solution

The vector $p(x) = 10x^2 - 17x + 9$ is in S if there exist real numbers c_1 and c_2 such that

$$c_1\left(x^2 - 2x + 3\right) + c_2\left(-2x^2 + 3x + 1\right) = 10x^2 - 17x + 9$$

Reorganizing the left side to collect common factors, we have

$$(c_1 - 2c_2)\,x^2 + (-2c_1 + 3c_2)\,x + (3c_1 + c_2) = 10x^2 - 17x + 9 \qquad (2)$$

The only way that two polynomials are equal is if coefficients of like terms are equal. Thus (2) is true only if there exist c_1 and c_2 that satisfy the system

$$\begin{array}{rcrcr} c_1 & - & 2c_2 & = & 10 \\ -2c_1 & + & 3c_2 & = & -17 \\ 3c_1 & + & c_2 & = & 9 \end{array}$$

Applying our usual solution methods, we can determine that the system has the unique solution $c_1 = 4$ and $c_2 = -3$. Hence $p(x)$ is in S.

Example 2

Let $S = \text{span}\left\{1, \cos(x), \cos(2x)\right\}$ be a subset of $C\ [0,\ \pi]$. Determine if $f(x) = \sin^2(x)$ is in S.

Solution

At first glance it may not appear that $f(x)$ is in S. However, recall from trigonometry the identity

$\sin^2(x) = \frac{1-\cos(2x)}{2}$

Hence it follows that $\sin^2(x)$ is a linear combination of two vectors in S, with

$$\sin^2(x) = \tfrac{1}{2}(1) - \tfrac{1}{2}\cos(2x)$$

Therefore $f(x) = \sin^2(x)$ is in S.

The preceding examples both involved spanning subsets S of a vector space. In Euclidean space $\mathbf{R}^n$, such sets are subspaces. The same is true in vector spaces.

THEOREM 6

Suppose that $\mathcal{V}$ is a (possibly infinite) subset of a vector space V, and let $S = \text{span}(\mathcal{V})$. Then S is a subspace of V.

The proof follows from verifying that the three conditions required of a subspace (Definition 3 in Lesson 1, Vector Spaces and Subspaces) are met. It is left as an exercise.

Example 3

Let $S = \text{span}\left\{\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 \\ 1 & 3 \end{bmatrix}, \begin{bmatrix} 4 & 1 \\ -2 & 1 \end{bmatrix}\right\}$ in $\mathbf{R}^{2\times 2}$ be a subspace of the vector space $\mathbf{R}^{2\times 2}$. Is $\mathbf{v} = \begin{bmatrix} 2 & 5 \\ -3 & 4 \end{bmatrix}$ in S?

Solution

In order for v to be in S, there must exist scalars c_1, c_2, and c_3 such that

$$\begin{array}{l} c_1\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} + c_2\begin{bmatrix} 0 & -1 \\ 1 & 3 \end{bmatrix} + c_3\begin{bmatrix} 4 & 1 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ -3 & 4 \end{bmatrix} \\ \Rightarrow \begin{bmatrix} c_1 & -c_2 + 4c_3 \\ (2c_1 + c_2 - 2c_3) & (c_1 + 3c_2 + c_3) \end{bmatrix} = \begin{bmatrix} 2 & 5 \\ -3 & 4 \end{bmatrix} \end{array}$$

For the components to be equal, c_1, c_2, and c_3 must satisfy the linear system

$$\begin{array}{rcrcr} c_1 & & + & 4c_3 & = & 2 \\ & - & c_2 & + & c_3 & = & 5 \\ 2c_1 & + & c_2 & - & 2c_3 & = & -3 \\ c_1 & + & 3c_2 & + & c_3 & = & 4 \end{array}$$

Our standard solution methods can be used to show that this system has no solutions, so v is not in S.

As we can see from our examples, even though vector spaces can be made up of polynomials, matrices, or other objects, answering questions about spanning sets often boils down to something that we have done again and again: solving a system of linear equations. But this is not always the case.

Example 4

Can a finite set of vectors span $\mathbf{P}$, the set of polynomials with real coefficients?

Solution

Suppose that $\{f_1(x),\ f_2(x), \ldots,\ f_m(x)\}$ is a set of polynomials in $\mathbf{P}$, sorted by degree with

$$\deg(f_1) \geq \deg(f_2) \geq \cdots \geq \deg(f_m)$$

Then any linear combination

$$c_1\ f_1(x) + c_2\ f_2(x) + \cdots + c_m f_m(x)$$

has degree at most that of $f_1(x)$ (and possibly less, if there is cancellation). Thus, if $n = \deg(f_1) + 1$, then $g(x) = x^n$ has degree greater than any linear combination of our set and hence cannot be in the span of the set. As there is nothing special about our set, this argument shows that no finite set of vectors can possibly span all of $\mathbf{P}$.

The argument in Example 4 that shows that a finite set cannot span $\mathbf{P}$ does not apply to the infinite set $\left\{1,\ x,\ x^2,\ x^3, \ldots\right\}$. This infinite set does span $\mathbf{P}$. Verification is left as an exercise.

Linear Independence

We now turn to the definition of linear independence. The definition is a near duplicate of the definition for vectors in Euclidean space.

DEFINITION 7

Let $\mathcal{V} = \{\mathbf{v}_1,\ \mathbf{v}_2, \ldots,\ \mathbf{v}_m\}$ be a set of vectors in a vector space V. Then $\mathcal{V}$ is linearly independent if the equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_m\mathbf{v}_m = 0$$

has only the trivial solution $c_1 = \cdots = c_m = 0$. The set $\mathcal{V}$ is linearly dependent if the equation has any nontrivial solutions.

As with span, the definition of linear independence extends to infinite sets. If $\mathcal{V}$ consists of infinitely many vectors, then $\mathcal{V}$ is linearly independent if every *finite* subset of $\mathcal{V}$ is linearly independent.

Example 5

Determine if

$$\left\{x^3 - 2x + 4,\ x^3 + 2x^2 - 2,\ 2x^3 + x^2 - 3x + 5\right\}$$

is a linearly independent subset of $\mathbf{P}^3$.

Solution

Just as in Euclidean space, we start by setting

$$c_1\left(x^3 - 2x + 4\right) + c_2\left(x^3 + 2x^2 - 2\right) + c_3\left(2x^3 + x^2 - 3x + 5\right) = 0$$

and then determine the values of c_1, c_2, and c_3 that satisfy the equation. Collecting common factors gives us the equivalent equation

$$(c_1 + c_2 + 2c_3)\,x^3 + (2c_2 + c_3)\,x^2 + (-2c_1 - 3c_3)\,x + (4c_1 - 2c_2 + 5c_3) = 0$$

Our polynomial is identically zero if and only if the coefficients are all zero. This will be true for any solution to the homogeneous system

$$\begin{array}{rcrcrcr} c_1 & + & c_2 & + & 2c_3 & = & 0 \\ & & 2c_2 & + & c_3 & = & 0 \\ -2c_1 & & & - & 3c_3 & = & 0 \\ 4c_1 & - & 2c_2 & + & 5c_3 & = & 0 \end{array}$$

Applying our standard solution methods shows that the system has infinitely many solutions, among them $c_1 = 3$, $c_2 = 1$, and $c_3 = -2$. Since a nontrivial linear combination of our vectors equals the zero vector, our set is linearly dependent.

Example 6

Determine if the subset

$$\left\{\begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 3 & 1 & 0 \\ 2 & 2 & 2 \end{bmatrix}\right\}$$

of $\mathbf{R}^{2\times 3}$ is linearly independent.

Solution

We proceed just as we did in the preceding example, by setting up the equation

$$c_1\begin{bmatrix} 1 & 0 & 2 \\ -1 & 1 & 0 \end{bmatrix} + c_2\begin{bmatrix} 3 & 1 & 0 \\ 2 & 2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Comparing the components on each side gives the linear system

$$\begin{array}{rcr} c_1 + 3c_2 & = & 0 \\ c_2 & = & 0 \\ 2c_1 & = & 0 \\ -c_1 + 2c_2 & = & 0 \\ c_1 + 2c_2 & = & 0 \\ 2c_2 & = & 0 \end{array}$$

We can see that this system has only the trivial solution $c_1 = c_2 = 0$, so the set is linearly independent.

A few observations about linear independence carry over from Euclidean space (proofs left as exercises):

- The set $\{0,\ \mathbf{v}_1, \cdots,\ \mathbf{v}_m\}$ is linearly dependent for all vectors $\mathbf{v}_1, \cdots,\ \mathbf{v}_m$.
- A set of two nonzero vectors $\{\mathbf{v}_1,\ \mathbf{v}_2\}$ is linearly dependent if and only if one is a scalar multiple of the other. (We could solve Example 6 using this fact.)
- A set with just one vector $\{\mathbf{v}_1\}$ is linearly dependent if and only if $\mathbf{v}_1 = 0$.

Example 7

Determine if $\left\{1,\ x,\ x^2,\ x^3, \ldots\right\}$ is a linearly independent subset of $\mathbf{P}$, the set of all polynomials with real coefficients.

Solution

Here we are working with an infinite set, so we must show that every finite subset is linearly independent. Let's take a typical finite subset

$$\left\{x^{a_1},\ x^{a_2}, \ldots,\ x^{a_m}\right\}$$

where $a_1,\ a_2, \ldots,\ a_m$ are distinct nonnegative integers. Suppose that

$$c_1x^{a_1} + c_2x^{a_2} + \cdots + c_mx^{a_m} = 0$$

We have the trivial solution $c_1 = c_2 = \cdots = c_m = 0$, but are there others? The answer is no, because a polynomial is identically zero (that is, zero for all x) only if all the coefficients are zero. Since we have only the trivial solution, our subset is linearly independent. Since our subset is arbitrary, it follows that all finite subsets are linearly independent and therefore our original infinite set is linearly independent.

Example 8

Determine if $\left\{1,\ x,\ e^x\right\}$ is a linearly independent subset of $C\ [0,\ 10]$.

Solution

Just as in the other examples, we ask ourselves if there exist nontrivial scalars c_1, c_2, and c_3 such that

$$c_1(1) + c_2x + c_3e^x = 0 \qquad (3)$$

An equivalent formulation is

$$c_1 + c_2x = -c_3e^x$$

Note that regardless of choice for c_3 and c_2, the left side is a linear equation. Since e^x is an exponential function, the only way the right side is linear is if $c_3 = 0$. This in turn forces $c_1 = c_3 = 0$ as well, which shows that the only solution to our original equation is the trivial one. Therefore our set is linearly independent.

In Example 8 we used the fact that e^x is not a linear function. But suppose we did not know that? Let's look at this example again, this time using a different approach.

Example 9

Determine if $\left\{1,\ x,\ e^x\right\}$ is a linearly independent subset of $C\ [0,\ 10]$.

Solution

Suppose that c_1, c_2, and c_3 satisfy (3). Then the equation is satisfied for every value of x, and hence in particular for each of $x = 0, x = 1$, and $x = 2$. Setting x equal to each of these values and plugging into (3) yields the homogeneous linear system

$$\begin{array}{rcrcrcr} c_1 & & & + & c_3 & = & 0 \\ c_1 & + & c_2 & + & ec_3 & = & 0 \\ c_1 & + & 2c_2 & + & e^2c_3 & = & 0 \end{array} \qquad (4)$$

By applying our standard methods, we can verify that the system (4) has only the trivial solution $c_1 = c_2 = c_3 = 0$. Hence the only possible solution to (3) is also the trivial one, because any nontrivial solution to (3) would also satisfy (4). Therefore the set $\left\{1,\ x,\ e^x\right\}$ is linearly independent.

A note of warning: The method in Example 9 cannot be used to show that a set of functions is linearly dependent. To see why, suppose that $f(x) = -x^3 + 3x^2 - 2x$ and $g(x) = \sin(2\pi x)$. Then

$$f(0) = g(0),\ f(1) = g(1),\ f(2) = g(2)$$

So if we apply the approach in Example 9, we get a linear system

$$\begin{array}{l} c_1 + c_2 = 0 \\ c_1 + c_2 = 0 \\ c_1 + c_2 = 0 \end{array}$$

Clearly this system has nontrivial solutions. However, the functions $f(x)$ and $g(x)$ are not multiples of each other (see Figure 1) and so are linearly independent. Functions as vectors are linearly dependent only if the dependence relation holds for all domain values, not just a few isolated values.

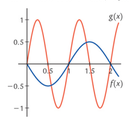


Figure 1 $f(x) = -x^3 + 3x^2 - 2x$ and $g(x) = \sin(2\pi x)$.

Span and Linear Independence

Sometimes it is difficult to keep straight the concepts of span and linear independence. The last two theorems in this lesson highlight the distinction between the two concepts.

THEOREM 8

The set $\mathcal{V} = \{\mathbf{v}_1,\ \mathbf{v}_2, \ldots,\ \mathbf{v}_m\}$ of nonzero vectors is linearly dependent if and only if at least one vector in the set is in the span of the others.

The proof of Theorem 8 is left as an exercise.

Example 10

Determine if the set of vectors

$$\left\{1, \cos(x), \cos(2x), \sin^2(x)\right\} \qquad (5)$$

is a linearly independent subset of $C\ [0,\ 10]$.

Solution

In Example 2 we showed that $\sin^2(x)$ is in $\text{span}\left\{1, \cos(x), \cos(2x)\right\}$. Hence by Theorem 8 the set of vectors in (5) is linearly dependent.

THEOREM 9

Let $\mathcal{V} = \{\mathbf{v}_1,\ \mathbf{v}_2, \ldots,\ \mathbf{v}_m\}$ be a subset of a vector space V. Then:

(a) The set $\mathcal{V}$ is *linearly independent* if and only if the equation $c_1\mathbf{v}_1 + \cdots + c_m\mathbf{v}_m = \mathbf{v}$ has *at most* one solution for each $\mathbf{v}$ in V.

(b) The set $\mathcal{V}$ spans V if and only if the equation $c_1\mathbf{v}_1 + \cdots + c_m\mathbf{v}_m = \mathbf{v}$ has *at least* one solution for each $\mathbf{v}$ in V.

The proof is left as an exercise. Note that Theorem 9 applies to subspaces as well as vector spaces.

Example 11

Let S be the subspace of $T\ (2,\ 2)$ of linear transformations of the form

$$T\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} a_1x_1 \\ a_2x_2 \end{bmatrix} \qquad (6)$$

where a_1 and a_2 are scalars. Suppose that

$$T_1\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} \text{ and } T_2\left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ x_2 \end{bmatrix}$$

Show that the set $\{T_1,\ T_2\}$ is linearly independent and spans S.

Solution

Suppose that T is a linear transformation in S and hence has the form in (6). To apply Theorem 9, we need to determine the number of solutions to the equation $c_1T_1(\mathbf{x}) + c_2T_2(\mathbf{x}) = T(\mathbf{x})$. This equation is equivalent to

$$c_1\begin{bmatrix} x_1 \\ 0 \end{bmatrix} + c_2\begin{bmatrix} 0 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_1x_1 \\ a_2x_2 \end{bmatrix}$$

For this to hold for all x_1 and x_2, we must have $c_1 = a_1$ and $c_2 = a_2$. Thus the equation $c_1T_1(\mathbf{x}) + c_2T_2(\mathbf{x}) = T(\mathbf{x})$ has exactly one solution for any a_1 and a_2. Therefore by Theorem 9(a) the set $\{T_1,\ T_2\}$ is linearly independent, and by Theorem 9(b) the set $\{T_1,\ T_2\}$ spans S.

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